

VisioCodicis: Disaster Routing Simulation

Generated: 2026-05-16 13:45:41

Routing area: Bulacan, Philippines | A* vs. ABSRA (Arc-flag Bidirectional Search A*)

1. Baseline Synthetic Disaster Scenarios

Lvl	Start -> Goal	Algo	Cost (min)	Nodes	Time (ms)
L1 Minor	HQ_Malolos -> Hagonoy	A*	17.45	4	0.009
L1 Minor	HQ_Malolos -> Hagonoy	ABSRA	17.45	2	0.013
L1 Minor	HQ_Malolos -> San Miguel (Viola St)	A*	79.56	15	0.021
L1 Minor	HQ_Malolos -> San Miguel (Viola St)	ABSRA	79.56	3	0.012
L1 Minor	HQ_Malolos -> Bocaue (Crossing)	A*	21.33	8	0.013
L1 Minor	HQ_Malolos -> Bocaue (Crossing)	ABSRA	21.33	4	0.013
L1 Minor	HQ_Malolos -> Calumpit (Market)	A*	29.02	10	0.016
L1 Minor	HQ_Malolos -> Calumpit (Market)	ABSRA	29.02	3	0.012
L1 Minor	Plaridel -> Bocaue	A*	19.64	6	0.012
L1 Minor	Plaridel -> Bocaue	ABSRA	19.64	3	0.013
L2 Moderate	HQ_Malolos -> Hagonoy	A*	32.14	8	0.015
L2 Moderate	HQ_Malolos -> Hagonoy	ABSRA	32.14	3	0.013
L2 Moderate	HQ_Malolos -> San Miguel (Viola St)	A*	59.95	12	0.019
L2 Moderate	HQ_Malolos -> San Miguel (Viola St)	ABSRA	59.95	3	0.012
L2 Moderate	HQ_Malolos -> Bocaue (Crossing)	A*	29.36	10	0.018
L2 Moderate	HQ_Malolos -> Bocaue (Crossing)	ABSRA	29.36	4	0.014
L2 Moderate	HQ_Malolos -> Calumpit (Market)	A*	46.18	13	0.019
L2 Moderate	HQ_Malolos -> Calumpit (Market)	ABSRA	46.18	3	0.012
L2 Moderate	Plaridel -> Bocaue	A*	39.09	9	0.016
L2 Moderate	Plaridel -> Bocaue	ABSRA	39.09	4	0.014
L3 Severe	HQ_Malolos -> Hagonoy	A*	27.41	6	0.010
L3 Severe	HQ_Malolos -> Hagonoy	ABSRA	27.41	2	0.011
L3 Severe	HQ_Malolos -> San Miguel (Viola St)	A*	180.57	10	0.014
L3 Severe	HQ_Malolos -> San Miguel (Viola St)	ABSRA	180.57	3	0.011
L3 Severe	HQ_Malolos -> Bocaue (Crossing)	A*	INF	10	0.013
L3 Severe	HQ_Malolos -> Bocaue (Crossing)	ABSRA	INF	2	0.008

L3 Severe	HQ_Malolos -> Calumpit (Market)	A*	27.45	7	0.011
L3 Severe	HQ_Malolos -> Calumpit (Market)	ABSRA	27.45	3	0.011
L3 Severe	Plaridel -> Bocaue	A*	INF	10	0.013
L3 Severe	Plaridel -> Bocaue	ABSRA	INF	2	0.009

2. Flood Scenarios (Mamuyac 2025 threshold: 25 cm | NDRRMC Carina Jul 2024)

Lvl	Start -> Goal	MaxDepth(cm)	>25cm edges	Closed roads	A* (min)	ABSRA (min)	A* Nodes	ABSRA Nodes	A* Time(ms)	ABSRA Time(ms)
L1	HQ_Malolos -> Hagonoy	21.1	0	0	12.77	12.77	4	2	0.008	0.012
L1	HQ_Malolos -> San Miguel (Viola St)	21.1	0	0	58.43	58.43	15	3	0.019	0.012
L1	HQ_Malolos -> Bocaue (Crossing)	21.1	0	0	21.79	21.79	10	4	0.016	0.013
L1	HQ_Malolos -> Calumpit (Market)	21.1	0	0	29.11	29.11	10	3	0.015	0.012
L1	Plaridel -> Bocaue	21.1	0	0	20.03	20.03	6	3	0.011	0.012
L2	HQ_Malolos -> Hagonoy	64.2	8	1	24.34	24.34	4	2	0.008	0.013
L2	HQ_Malolos -> San Miguel (Viola St)	64.2	8	1	74.92	74.92	14	3	0.021	0.012
L2	HQ_Malolos -> Bocaue (Crossing)	64.2	8	1	44.06	44.06	11	4	0.017	0.013
L2	HQ_Malolos -> Calumpit (Market)	64.2	8	1	INF	INF	14	2	0.019	0.009
L2	Plaridel -> Bocaue	64.2	8	1	25.98	25.98	6	3	0.011	0.013
L3	HQ_Malolos -> Hagonoy	126.2	15	11	INF	INF	4	2	0.006	0.008
L3	HQ_Malolos -> San Miguel (Viola St)	126.2	15	11	INF	INF	4	1	0.006	0.007
L3	HQ_Malolos -> Bocaue (Crossing)	126.2	15	11	INF	INF	4	2	0.006	0.008
L3	HQ_Malolos -> Calumpit (Market)	126.2	15	11	INF	INF	4	1	0.006	0.007
L3	Plaridel -> Bocaue	126.2	15	11	INF	INF	1	1	0.002	0.006

3. Heavy-Traffic Scenarios (BPR model | DPWH Region III / TomTom 2024)

Lvl	Start -> Goal	Max V/C	Edges >= 1.0	A* (min)	ABSRA (min)	A* Nodes	ABSRA Nodes	Blocked roads	A* Time(ms)	ABSRA Time(ms)
L1	HQ_Malolos -> Hagonoy	1.00	9	12.72	12.72	4	2	0	0.008	0.012
L1	HQ_Malolos -> San Miguel (Viola St)	1.00	9	58.43	58.43	15	3	0	0.020	0.012
L1	HQ_Malolos -> Bocaue (Crossing)	1.00	9	22.56	22.56	10	4	0	0.017	0.013
L1	HQ_Malolos -> Calumpit (Market)	1.00	9	28.27	28.27	10	3	0	0.016	0.012
L1	Plaridel -> Bocaue	1.00	9	21.12	21.12	8	3	0	0.014	0.012
L2	HQ_Malolos -> Hagonoy	1.28	15	15.09	15.09	4	2	0	0.008	0.012
L2	HQ_Malolos -> San Miguel (Viola St)	1.28	15	69.38	69.38	15	3	0	0.019	0.012
L2	HQ_Malolos -> Bocaue (Crossing)	1.28	15	23.47	23.47	9	4	0	0.016	0.013

L2	HQ_Malolos -> Calumpit (Market)	1.28	15	33.39	33.39	11	3	0	0.016	0.012
L2	Plaridel -> Bocaue	1.28	15	23.55	23.55	6	3	0	0.011	0.012
L3	HQ_Malolos -> Hagonoy	1.72	15	18.94	18.94	4	2	0	0.007	0.012
L3	HQ_Malolos -> San Miguel (Viola St)	1.72	15	105.22	105.22	15	3	0	0.019	0.012
L3	HQ_Malolos -> Bocaue (Crossing)	1.72	15	37.03	37.03	11	4	0	0.016	0.013
L3	HQ_Malolos -> Calumpit (Market)	1.72	15	46.60	46.60	11	3	0	0.016	0.012
L3	Plaridel -> Bocaue	1.72	15	37.07	37.07	8	3	0	0.014	0.012

4. Road-Block Scenarios (DPWH Region III | Bulacan DEO post-typhoon bulletins)

Lvl	Start -> Goal	Max S	Edges >= 0.6	Closed roads	A* (min)	ABSRA (min)	A* Nodes	ABSRA Nodes	A* Time(ms)	ABSRA Time(ms)
L1	HQ_Malolos -> Hagonoy	0.28	0	0	18.38	18.38	4	2	0.007	0.012
L1	HQ_Malolos -> San Miguel (Viola St)	0.28	0	0	58.43	58.43	15	3	0.020	0.012
L1	HQ_Malolos -> Bocaue (Crossing)	0.28	0	0	22.76	22.76	10	4	0.016	0.013
L1	HQ_Malolos -> Calumpit (Market)	0.28	0	0	26.59	26.59	10	3	0.015	0.012
L1	Plaridel -> Bocaue	0.28	0	0	20.91	20.91	8	3	0.014	0.012
L2	HQ_Malolos -> Hagonoy	0.54	0	0	21.18	21.18	4	2	0.007	0.012
L2	HQ_Malolos -> San Miguel (Viola St)	0.54	0	0	103.93	103.93	15	3	0.019	0.012
L2	HQ_Malolos -> Bocaue (Crossing)	0.54	0	0	33.45	33.45	10	4	0.016	0.015
L2	HQ_Malolos -> Calumpit (Market)	0.54	0	0	51.25	51.25	13	3	0.019	0.012
L2	Plaridel -> Bocaue	0.54	0	0	23.00	23.00	6	3	0.011	0.012
L3	HQ_Malolos -> Hagonoy	0.67	10	10	INF	INF	5	2	0.007	0.009
L3	HQ_Malolos -> San Miguel (Viola St)	0.67	10	10	INF	INF	5	2	0.007	0.008
L3	HQ_Malolos -> Bocaue (Crossing)	0.67	10	10	INF	INF	5	2	0.007	0.008
L3	HQ_Malolos -> Calumpit (Market)	0.67	10	10	INF	INF	5	2	0.007	0.008
L3	Plaridel -> Bocaue	0.67	10	10	INF	INF	5	2	0.007	0.008

5. Algorithm Comparison Summary (averaged per scenario type & level)

Scenario	Level	Algorithm	Avg Cost (min)	Avg Nodes	Avg Time (ms)	Reachable
Generic	L1	A*	33.40	8.6	0.014	5/5
Generic	L1	ABSRA	33.40	3.0	0.013	5/5
Generic	L2	A*	41.34	10.4	0.017	5/5
Generic	L2	ABSRA	41.34	3.4	0.013	5/5
Generic	L3	A*	78.48	7.7	0.012	3/5
Generic	L3	ABSRA	78.48	2.7	0.011	3/5
Flood	L1	A*	28.43	9.0	0.014	5/5
Flood	L1	ABSRA	28.43	3.0	0.012	5/5
Flood	L2	A*	42.33	8.8	0.014	4/5
Flood	L2	ABSRA	42.33	3.0	0.013	4/5
Flood	L3	A*	-	-	-	0/5
Flood	L3	ABSRA	-	-	-	0/5
Traffic	L1	A*	28.62	9.4	0.015	5/5
Traffic	L1	ABSRA	28.62	3.0	0.012	5/5
Traffic	L2	A*	32.98	9.0	0.014	5/5
Traffic	L2	ABSRA	32.98	3.0	0.012	5/5
Traffic	L3	A*	48.97	9.8	0.015	5/5
Traffic	L3	ABSRA	48.97	3.0	0.012	5/5
Road-Block	L1	A*	29.41	9.4	0.014	5/5
Road-Block	L1	ABSRA	29.41	3.0	0.012	5/5
Road-Block	L2	A*	46.56	9.6	0.015	5/5
Road-Block	L2	ABSRA	46.56	3.0	0.012	5/5
Road-Block	L3	A*	-	-	-	0/5
Road-Block	L3	ABSRA	-	-	-	0/5